CHEMISTRY ONLY

# 27 CARBOXYLIC ACIDS

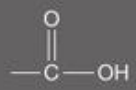
Previously you have seen how alcohols can be oxidised to form a class of organic compounds called carboxylic acids. Here we look at their structures and some of their reactions in more detail.



▲ Figure 27.1 Stinging nettles inject methanoic acid.

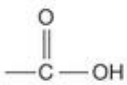
## LEARNING OBJECTIVES

- Know that carboxylic acids contain the functional group



- Understand how to draw structural and displayed formulae for unbranched-chain carboxylic acids with up to four carbon atoms in the molecule, and name each compound
- Describe the reactions of aqueous solutions of carboxylic acids with metals and metal carbonates
- Know that vinegar is an aqueous solution containing ethanoic acid

Acids such as ethanoic acid are known as **carboxylic acids** and they all contain the functional group **–COOH**



The most familiar one is ethanoic acid (old name *acetic acid*). **Vinegar** is *an aqueous solution containing ethanoic acid*.

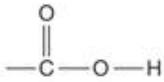
Carboxylic acids are formed by oxidation of the corresponding alcohols, for example ethanoic acid is formed when ethanol is left in the air or oxidised using potassium dichromate(VI) in dilute sulfuric acid. If you need a reminder, have at look at Chapter 26, pages 289–290.

## DRAWING AND NAMING THE CARBOXYLIC ACIDS

Again, we are only going to be looking at the four simplest carboxylic acids: methanoic acid, ethanoic acid, propanoic acid and butanoic acid. These are shown in Table 27.1.

**HINT**

If you are asked for a displayed formula in an exam, you should draw all the bonding in the COOH group as shown here. If you just want to write the formula in an equation, it is normal to write it as, for example, $CH_3COOH$.

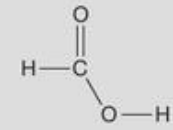
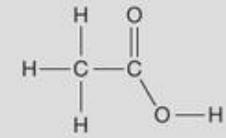
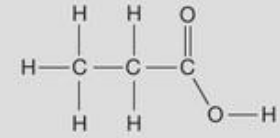
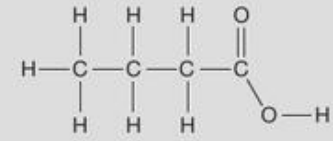


▲ Figure 27.2 The carboxylic acid functional group

**KEY POINT**

It is really important to notice that when we name a carboxylic acid we count the total number of carbon atoms, including the one in the –COOH group. So propanoic acid has three carbon atoms in total, which is why it starts with 'prop'. The –COOH group is always on the end carbon, so there is no need to insert a number in the name to represent the position of the functional group.

Table 27.1 Formulae and names of the first four members of the carboxylic acid homologous series

| Name | Molecular formula | Structural formula | Displayed formula |
|---|---|---|---|
| methanoic acid | $CH_2O_2$ | HCOOH |  |
| ethanoic acid | $C_2H_4O_2$ | $CH_3COOH$ |  |
| propanoic acid | $C_3H_6O_2$ | $CH_3CH_2COOH$ |  |
| butanoic acid | $C_4H_8O_2$ | $CH_3CH_2CH_2COOH$ |  |

## ACID PROPERTIES OF THE CARBOXYLIC ACIDS

Ethanoic acid and the other carboxylic acids are weak acids with a pH of about 3–5, depending on the concentration of the solution. They will turn blue litmus paper red, and react with all the things you expect acids to react with.

In the reactions described here, we are taking ethanoic acid as an example because it is the one you are most likely to use. All the others will behave similarly, and if necessary all you need to do is change the formula from $CH_3COOH$ to, for example, $CH_3CH_2COOH$.

### REACTION WITH METALS



▲ Figure 27.3 Reaction between magnesium ribbon and ethanoic acid.

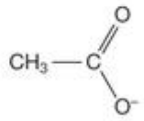
Dilute ethanoic acid reacts with metals in the same way as other dilute acids such as hydrochloric acid or sulfuric acid, only *more slowly*. Remember *metals react with acids to give a salt and hydrogen* (see Chapter 17, page 174).

For example, dilute ethanoic acid reacts with magnesium with a lot of fizzing to produce a salt and hydrogen. The salt formed is magnesium ethanoate, which is soluble in water and so you get a colourless solution:

$$Mg(s) + 2CH_3COOH(aq) \rightarrow (CH_3COO)_2Mg(aq) + H_2(g)$$

Students frequently have problems with the formulae of salts like magnesium ethanoate. You have to remember that this is an ionic compound, and so

contains magnesium ions, $Mg^{2+}$, and ethanoate ions, $CH_3COO^-$. That means that you need two ethanoate ions for every magnesium ion to make the charges balance. The metal is written after the ethanoate because that reflects the structure; the magnesium ion is attracted to the negative charge on the oxygen atoms in the ethanoate ions.



▲ Figure 27.4 The structure of an ethanoate ion

Other metals behave as you would expect. Sodium is dangerously reactive; zinc produces bubbles of hydrogen more slowly than it does with hydrochloric acid, for example; metals below hydrogen in the reactivity series won't react.

Wherever a reaction takes place, a metal ethanoate will be formed as well as hydrogen. Zinc ethanoate is $(CH_3COO)_2Zn$.

**EXTENSION WORK**

Ethanoic acid is a weak acid, whereas the acids we have met before, hydrochloric acid or sulfuric acid, are called strong acids. Students can have the misconception that strong acids are equivalent to concentrated acids. Whether an acid is strong or weak has, in fact, nothing to do with its concentration. Strong acids dissociate completely into their constituent ions in water, for example in hydrochloric acid you find mostly $H^+$ and $Cl^-$ ions and hardly any HCl molecules. In weak acids, the dissociation is only partial. About 1 in ~75 $CH_3COOH$ molecules will dissociate into $H^+$ and $CH_3COO^-$ in water. This is why if you have the same concentration of ethanoic acid and hydrochloric acid, the ethanoic acid reacts with metals or metal carbonates much more slowly than the hydrochloric acid as the concentration of the $H^+$ ions is much less.

## REACTION WITH CARBONATES

*Carbonates react with acids to give a salt, carbon dioxide and water* (Chapter 17, page 177), and ethanoic acid behaves like any other acid. For example, with sodium carbonate you get a lot of fizzing and a colourless solution of sodium ethanoate:

$$Na_2CO_3(s) + 2CH_3COOH(aq) \rightarrow 2CH_3COONa(aq) + CO_2(g) + H_2O(l)$$

With calcium carbonate you again get lots of fizzing and a colourless solution – this time containing calcium ethanoate:

$$2CH_3COOH(aq) + CaCO_3(s) \rightarrow (CH_3COO)_2Ca(aq) + CO_2(g) + H_2O(l)$$

## CHAPTER QUESTIONS

SKILLS INTERPRETATION

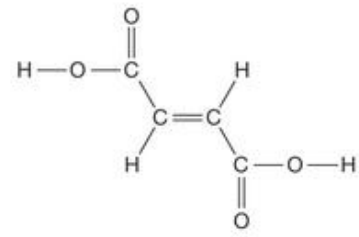
1 a Draw the structural formulae for the following compounds:
   i ethanoic acid
   ii butanoic acid

   b Draw the displayed formulae for these compounds:
   i methanoic acid
   ii propanoic acid

   c Name the following compounds:
   i $CH_3CH_2CH_2COOH$
   ii $CH_3COOH$

SKILLS CRITICAL THINKING (6)

2 What are the functional groups shown in the compound below (choose one answer from the table below)?



| | Alkene | Alcohol | Carboxylic acid |
|---|---|---|---|
| A | ✓ | ✓ | ✓ |
| B | ✓ | ✓ | ✗ |
| C | ✓ | ✗ | ✓ |
| D | ✗ | ✓ | ✓ |

SKILLS INTERPRETATION (8)

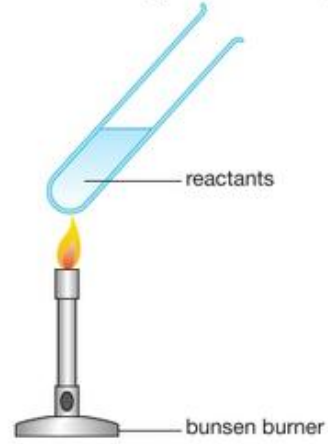
3 Propan-1-ol can be oxidised by heating it under reflux with potassium dichromate(VI) in dilute sulfuric acid.

a Draw the full displayed formula of the product of this reaction.

SKILLS CRITICAL THINKING
SKILLS EXECUTIVE FUNCTION (5)

b Describe the colour change during the reaction.

c A student suggested the following apparatus for the oxidation of propan-1-ol. Explain why this method might be too dangerous.




SKILLS CRITICAL THINKING, PROBLEM SOLVING (9)

d Describe what you would see and write balanced chemical equations for the reactions if you added some of the product you collected to test-tubes containing:

i magnesium ribbon ii sodium carbonate solution.

SKILLS PROBLEM SOLVING (8)

4 Vinegar can react with baking soda, which contains sodium hydrogencarbonate ($NaHCO_3$), to give a salt, carbon dioxide and water.

a Balance the following equation for the reaction between vinegar and baking soda.

$$\underset{\text{vinegar}}{_____} + \underset{\text{baking soda}}{NaHCO_3} \rightarrow \underset{\text{a salt}}{_____} + CO_2 + H_2O$$

SKILLS CRITICAL THINKING (6)

b Suggest the type of reaction occurring.

c Suggest an observation that could be made during this reaction.

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